

CHAPTER 1

THE LIVING WORLD



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1.1 Diversity in the Living World

1.2 Taxonomic Categories

Kitna amazing hai ye living world! Living beings ki variety really mind-blowing hai. Humein alag-alag habitats mein living organisms milte hain – chahe wo thande pahad ho, ghane jungle ho, samundar ho, fresh water lakes ho, deserts ho ya garam paani ke springs. Ek daudta hua ghoda, migrate karte birds ka lamba safar, phoolon se bhari valley ya shikaar karta shark – ye sab humein hairan kar dete hain aur nature ke liye ek deep sense of wonder feel hota hai. Ek population ke members ke beech ka ecological conflict aur cooperation, different populations ke beech ka relation, ya ek cell ke andar molecules ka lagataar movement – ye sab humein sochne par majboor karte hain: life aakhir hai kya? Is chapter mein do important questions chhipe hue hain. Pehla technical question hai – living aur non-living mein difference kya hai? Doosra philosophical question hai – life ka purpose kya hai? Scientists ke roop mein hum doosre question ka answer nahi denge. Is chapter mein hum pehle question par focus karenge – what is living?

1.1 DIVERSITY IN THE LIVING WORLD

Agar tum apne aas-paas dhyan se dekho, to tumhe bahut saare living organisms dikhenge – jaise potted plants, insects, birds, tumhare pets, aur aur bhi kai animals aur plants. Aise bhi kai organisms hote hain jo tumhe naked eye se nahi dikhte, lekin wo bhi tumhare around hi hote hain. Agar tum apni observations ka area badha do, to tumhe aur bhi zyada variety aur range ke organisms dekhne ko milenge. Obviously, agar tum kisi dense forest mein jaoge, to wahan tumhe living organisms ki sankhya aur unke kinds bahut zyada dikhenge. Har alag plant, animal ya organism jo tum dekhte ho, ek **species** ko represent karta hai. Ab tak jo species describe ki gayi hain, unki sankhya lagbhag 1.7 – 1.8 million ke beech hai. Isse hum **biodiversity** kehte hain, yani Earth par maujood organisms ki number aur variety.

types of organisms present on earth. We should remember here that as we explore new areas, and even old ones, new organisms are continuously being identified.

As stated earlier, there are millions of plants and animals in the world; we know the plants and animals in our own area by their local names. These local names would vary from place to place, even within a country. Probably you would recognise the confusion that would be created if we did not find ways and means to talk to each other, to refer to organisms we are talking about.

Hence, there is a need to standardise the naming of living organisms such that a particular organism is known by the same name all over the world. This process is called **nomenclature**. Obviously, nomenclature or naming is only possible when the organism is described correctly and we know to what organism the name is attached to. This is **identification**.

In order to facilitate the study, number of scientists have established procedures to assign a scientific name to each known organism. This is acceptable to biologists all over the world. For plants, scientific names are based on agreed principles and criteria, which are provided in International Code for Botanical Nomenclature (ICBN). You may ask, how are animals named? Animal taxonomists have evolved International Code of Zoological Nomenclature (ICZN). The scientific names ensure that each organism has only one name. Description of any organism should enable the people (in any part of the world) to arrive at the same name. They also ensure that such a name has not been used for any other known organism.

Biologists follow universally accepted principles to provide scientific names to known organisms. Each name has two components – the **Generic name** and the **specific epithet**. This system of providing a name with two components is called **Binomial nomenclature**. This naming system given by Carolus Linnaeus is being practised by biologists all over the world. This naming system using a two word format was found convenient. Let us take the example of mango to understand the way of providing scientific names better. The scientific name of mango is written as *Mangifera indica*. Let us see how it is a binomial name. In this name *Mangifera* represents the genus while *indica*, is a particular species, or a specific epithet. Other universal rules of nomenclature are as follows:

1. Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin.
2. The first word in a biological name represents the genus while the second component denotes the specific epithet.
3. Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin.

4. Genus ko denote karne wala pehla word hamesha capital letter se shuru hota hai, jabki specific epithet hamesha small letter se shuru hota hai. Isse hum *Mangifera indica* ke example se samajh sakte hain.

Author ka naam specific epithet ke baad aata hai, yani biological name ke end me likha jata hai aur ise abbreviated form me likha jata hai, jaise *Mangifera indica* Linn. Isse pata chalta hai ki is species ko sabse pehle Linnaeus ne describe kiya tha.

Kyunki itne saare living organisms ka study karna almost impossible hai, isliye isko possible banane ke liye kuch aise tareeke banane zaroori hain. Is process ko **classification** kehte hain. Classification wo process hai jisme kisi bhi cheez ko uske kuch easily observable characters ke basis par convenient categories me group kiya jata hai.

For example, hum easily plants ya animals ya dogs, cats ya insects jaise groups ko pehchan lete hain. Jab hum inme se koi term use karte hain, to hum us group ke kuch specific characters ko usse associate kar dete hain. Agar tum dog ke baare me sochte ho to obviously tumhe 'dogs' dikhte hain, 'cats' nahi. Ab, agar hum 'Alsations' ke baare me soche, to hume pata hota hai ki hum kis organism ki baat kar rahe hain. Isi tarah, agar hum 'mammals' bolte hain, to tum obviously un animals ke baare me sochoge jinke external ears hote hain aur body hair hoti hai. Waise hi, plants me agar hum 'Wheat' bolte hain, to hamare dimaag me wheat plant ki image aati hai, na ki rice ya koi aur plant. Isliye, 'Dogs', 'Cats', 'Mammals', 'Wheat', 'Rice', 'Plants', 'Animals', etc., aise convenient categories hain jo hum organisms ko study karne ke liye use karte hain. In categories ke liye scientific term **taxa** hai. Yahan tumhe samajhna hoga ki taxa bahut different levels par categories ko indicate kar sakte hain. Plants bhi ek taxa banate hain. Wheat bhi ek taxa hai. Isi tarah, 'animals', 'mammals', 'dogs' sab taxa hain – lekin tum jaante ho ki dog ek mammal hai aur mammals animals hote hain. Isliye, 'animals', 'mammals' aur 'dogs' different levels par taxa ko represent karte hain.

Isliye, characters ke basis par, saare living organisms ko different taxa me classify kiya ja sakta hai. Is process ko **taxonomy** kehte hain. External aur internal structure, saath hi cell ki structure, development process aur ecological information of organisms modern taxonomic studies ka essential hissa hain aur inhi par ye studies based hoti hain.

Isliye, characterisation, identification, classification aur nomenclature wo processes hain jo taxonomy ke liye basic hain.

Taxonomy koi nayi cheez nahi hai. Humans hamesha se hi alag-alag tarah ke organisms ke baare me zyada se zyada janne me interested rahe hain, khaaskar unke apne use ke liye. Purane samay me, humans ko apni basic needs jaise food, clothing aur shelter ke liye sources dhoondhne hote the. Isliye, shuruati classifications alag-alag organisms ke 'uses' ke basis par ki gayi thi.

Humans, tab se, sirf alag-alag organisms ke baare me aur unki diversity ke baare me hi nahi, balki unke beech ke relationships ke baare me bhi jaanne me interested rahe hain. Is branch of study ko **systematics** kehte hain. Systematics word Latin word 'systema' se aaya hai, jiska matlab hota hai systematic arrangement of organisms. Linnaeus

used *Systema Naturae* ko hi apne publication ke title ke roop mein use kiya. Baad mein, usne systems ko include karne ke liye identification, nomenclature aur classification ko enlarge kiya. Systematics ko organisms ke beech evolutionary relationships ko samajhne ke liye account mein liya gaya.

1.2 Taxonomic Categories

Classification ek single step process nahi hai, balki ismein har ek step mein ek hierarchy hoti hai, jismein har step ek rank ya category ko represent karta hai. Since ye category overall taxonomic arrangement ka part hai, ise **taxonomic category** kaha jata hai aur ye milkar **taxonomic hierarchy** ko constitute karta hai. Har category, classification ki ek unit hoti hai, jo rank ko represent karti hai aur ise **taxon** (pl.: taxa) ke naam se commonly term kiya jata hai.

Taxonomic categories aur hierarchy ko ek example se samjha sakte hain. Insects ek group of organisms hain, jo common features share karte hain, jaise teen pairs jointed legs. Iska matlab insects ko recognizable concrete objects mana ja sakta hai aur isliye inhe ek rank ya category di ja sakti hai. Kya aap in organisms ke aur groups ka naam rakh sakte hain? Yaad rakhiye, groups represent category. Category aur rank ko further denote karta hai. Har rank ya taxon, asl mein ek unit of classification hota hai. In taxonomic groups/categories mein distinct biological entities hoti hain aur ye sirf morphological aggregates nahi hote.

Sabhi jaane-maane organisms par taxonomic studies ne common categories jaise kingdom, phylum ya division (plants ke liye), class, order, family, genus aur species ka development kiya hai. Sabhi organisms, jinme plant aur animal kingdoms shamil hain, ko species se lekar lowest category tak divide kiya gaya hai. Ab aap soch sakte hain ki kisi organism ko different categories mein place karne ka basic requirement kya hai? Individual ya group of organisms ke characters ki knowledge is kaam mein help karti hai aur ye similarities aur dissimilarities ko identify karne mein madad karti hai, same kind ke organisms ke saath-saath other kinds ke organisms ke beech.

1.2.1 Species

Taxonomic studies mein fundamental similarities ko consider karte hue, individual organisms ke ek group ko **species** kaha jata hai. Kisi doosre closely related species se ek species ko distinguish karne mein ye help karta hai. Chaliye, ab hum *Mangifera indica*, *Solanum tuberosum* (potato) aur *Panthera leo* (lion) ko consider karte hain. Inke teen naam hain — *indica*, *tuberosum* aur *leo*, jo specific epithets ko represent karte hain, jabki first words *Mangifera*, *Solanum* aur *Panthera* genus ko represent karte hain. Har genus mein ek ya zyada species ho sakti hain, lekin har ek species alag-alag morphological similarities ko represent karti hai. Example ke liye, *Panthera* ka ek aur specific epithet hai *tigris* aur *Solanum* mein *Solanum* species jaise *nigrum* aur

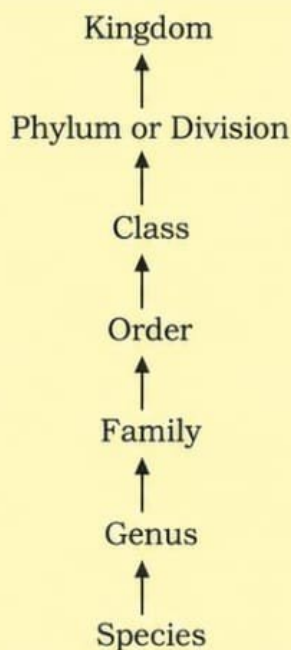


Figure 1.1 Taxonomic categories showing hierarchical arrangement in ascending order

in ke common features ke basis par, jaise notochord aur dorsal hollow neural system ki presence, phylum Chordata mein include kiya jata hai. Plants ke case mein, kuch similar characters wali classes ko ek higher category Division mein rakha jata hai.

1.2.7 Kingdom

Different phyla se belong karne wale sabhi animals ko animals ke classification system mein highest category Kingdom Animalia mein rakha jata hai. Dusri taraf, Kingdom Plantae alag hai aur ismein different divisions ke saare plants include hote hain. Isliye, hum in dono groups ko animal kingdom aur plant kingdom ke naam se refer karenge.

Species se kingdom tak ki taxonomic categories ko Figure 1.1 mein ascending order mein dikhaya gaya hai, jismein species se starting hoti hai. Ye broad categories hain. Lekin, taxonomists ne is hierarchy mein sub-categories bhi develop ki hain, taaki different taxa ko aur zyada proper aur scientific way mein place kiya ja sake.

Figure 1.1 mein di gayi hierarchy ko dekhiye. Kya aap arrangement ka basis yaad kar sakte hain? Jaise-jaise hum species se kingdom ki taraf higher category mein jaate hain, common characteristics ki number decrease hoti jaati hai. Taxa jitna lower hota hai, us taxon ke members mein utni hi zyada characteristics common hoti hain. Higher category par, same level ke doosre taxa ke saath relationship determine karna utna hi difficult hota hai. Isliye, classification ki problem aur complex ho jaati hai.

Table 1.1 kuch common organisms jaise housefly, man, mango aur wheat ki taxonomic categories ko show karti hai.

TABLE 1.1 Organisms with their Taxonomic Categories

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledonae	Angiospermae

SUMMARY

Duniya lots of living cheezo se bhari hui hai. Crores plants aur animals identify kiye gaye hain aur describe kiye gaye hain, lekin abhi bhi ek badi sankhya mein unknown cheezein bani hui hain. Size, colour, habitat, physiological aur morphological features ke terms mein in organisms ki bahut variety hoti hai, aur inke living organisms ki defining characteristics ko samajhne ke liye hume in features ko dekhna padta hai. Kisi bhi organism ki study ko facilitate karne ke liye, biologists ne kuch basic rules aur principles develop kiye hain, jo identification, nomenclature aur classification of organisms ke liye help karte hain. In aspects se judi knowledge ka branch, jise taxonomy kehte hain, unhi aspects ko refer karta hai.

Plant aur animal dono kingdoms ke species ki taxonomy ka study, agriculture, forestry, industry aur general field mein hamare bio-resources ko jaane ke liye bahut useful hota hai aur unki diversity ko samajhne mein madad karta hai. Taxonomy ke basic principles jaise identification, naming aur classification of organisms, universally international codes ke under evolve kiye gaye hain. Resemblances aur distinct differences ke basis par, har organism ko identify kiya jata hai aur usse correct scientific/biological name diya jata hai, jo binomial system of nomenclature ka part hota hai. Ek organism ko classification system mein ek jagah par rakhna ya uski position dikhana, taxonomy ki categories ya taxa ke generally referred ranks ko represent karta hai. Saare categories ek taxonomic hierarchy constitute karte hain.

EXERCISES

1. Living organisms ko classify karne ki zarurat kyu hoti hai?
2. Classification system har waqt badalti rehti hai, abhi kyu aur kaise?
3. Aap kaunse criteria choose karenge taaki aise logon ko classify kar sakein jinse aap aksar milte hain?
4. Aapne individuals aur populations ki identification se kya seekha?
5. Mango ka scientific name kya hai? Sahi tarike se likha hua naam identify karo.
Mangifera Indica
Mangifera indica
6. Ek taxon ko define karo. Kuch examples do jo different hierarchical levels par taxa ke hote hain.
7. Kya aap sahi taxonomical category ka sequence identify kar sakte ho?
(a) Species → Order → Phylum → Kingdom
(b) Genus → Species → Order → Kingdom
(c) Species → Genus → Order → Phylum
8. Try karo ki ab tak jo saare current accepted meanings hain word 'species' ke, unhe collect karo. Apne teacher ke saath discuss karo ki higher plants aur animals mein se kaunse plants aur bacteria ek doosre haath par hote hain.
9. In terms ko define aur samjhao:
(i) Phylum (ii) Class (iii) Family (iv) Order (v) Genus
10. Ek plant aur ek animal ka taxonomical hierarchy ko suitable examples ke saath illustrate karo.



PROJECT
DHA

प्रज्ञा

चेत

गेआन